Xiyao Long Paper response

Spanner: Google's globally-distributed database

Distributed data-management has always been done to make replicas of data across different physical locations/devices for data security and reducing latency. This paper talks about a distributed data storing infrastructure called "spanner". This infrastructure is superior to some other distributed data systems in a number of ways. It can support storing data that are complex and muti-versioned, it also allows the data-management to be controlled by individual applications. But most importantly, the spanner guarantee that globally any transaction committed in logical/physical time would have a timestamp than those of transactions that happened later. This is enabled by that Google implemented their global cluster management system that can track time very accurately using a combination of GPS and atomic clocks. When two seemingly concurrent transactions happen, the system can wait out in a relatively short time(bound) because of the global cluster system.

The implementation of spanner involves using a data-replication hierarchy consists of universemaster, zonemaster, zone servers. The universemaster oversee zonemasters and direct replicating data between zones very few minutes. The zonemaster controls many spanservers which serve data to clients. The clients are able to locate their corresponding spanservers. In each zone, spanservers identities among themselves a leader. Each spanserver has several hundred instances of a tablet with its corresponding state machine. A bag of Key-value mappings is consistently replicated among the state machines, therefore, allow the identification of a leader. The spanserver would use a lock table on the leader to enable synchronization of transactions between leaders of different spanservers. Every replicated data can decide whether its served data is up-to-date enough for a data read based on if the "read time stamp" is smaller than "maximum up-to-date time-stamp".

I think the really innovative solution in this paper should be the Paxos groups in which a leader would have its data synchronized with those of leaders from other groups. This allows balancing data loads between servers and allows scaling up the system with more and more machines. Another point is how up-to-date data needs to be for a client reading can also be tuned based on needs through the safe time for Paxos state machine and safe time for transaction manager, therefore, make the system more flexible.

As for the insufficiency of this paper, I think they can explain what are the leaders in Paxos groups more clearly. And they can explain the entire process of synchronization/replication rather than giving a coarse-grained explanation and going to details latter. They give several metrics for evaluation but I think it also helps if they can explain in general why they use those metrics.